

OpTC PPT-aligned reproducibility and reviewer audit

Frozen chronological protocol; no provider calls were made during this packaging pass.

Raw telemetry	16,902,846 events
Online stream	3,002 chronological checkpoints
Frozen matrix	3,002 x 5 = 15,010 strict-schema-valid responses
Official reconstruction	76 observable red-team steps and 73 official adjacent transitions across 3 campaigns
Auxiliary detection	207 non-overlapping host-local 30-minute windows: 35 attack and 172 background
CaMVo operating point	delta=0.80, k_min=3, transferred unchanged from the frozen Mordor protocol

Task definitions

The three units are intentionally not added together. The 3,002 checkpoints measure online routing, convergence, latency, and cost. The 76 official steps and 73 adjacent transitions measure red-team-step recall, timeline reconstruction, and attack-path continuity. The 207 windows provide an auxiliary attack/background detection test. The 15,010 responses are paired model outputs, not 15,010 independent ground-truth test examples.

Evidence and reachability

Official labels remain evaluator-only. Of 101 documented red-team steps, 76 have observable telemetry in the downloaded scope. A frozen model-blind alignment requires exact campaign/day, host, actor_id, and object_id identity with official positive evidence within the 30-minute scale selected in training folds. It makes 52/76 steps prompt-reachable. CaMVo and online full voting recover 49/52 reachable steps, which is reported together with the end-to-end value 49/76. The exact-event sensitivity audit is 16/17 reachable and 16/76 end-to-end.

Time and leakage controls

Prompts contain only the current checkpoint and strictly past graph state. Model selection is completed before current responses are observed. Router reliability uses only prior assigned aggregate consensus; it does not use current or future official labels. The online full-vote comparator begins from the same pre-response learned reliability state as CaMVo and queries all five models. No held-out campaign labels influence the current route.

Selection, uncertainty, and limitations

Nested temporal-scale selection

Candidate window widths are 1, 2, 3, 4, 5, 10, 15, and 30 minutes. For each held-out campaign, width and one common threshold are selected using only the other two campaigns; all three folds select 30 minutes. Every method in a fold receives the same width and threshold. The 1-minute evaluation (1,306 windows, 183 attack) is retained as sensitivity because it fragments multi-event chains and is not the selected primary scale.

Accuracy and imbalance

All 207 selected-scale windows remain in the auxiliary test. Balanced accuracy is the mean of attack recall and background recall, so the two classes contribute equally despite the natural 35:172 ratio. Ordinary fraction correct remains a secondary deployment audit. An always-background rule scores 50.0% balanced accuracy and detects 0/35 attacks; its 172/207 ordinary fraction is not an algorithmic baseline.

Uncertainty

Window metrics use campaign-aware paired bootstrap intervals. Official-step recall includes Wilson intervals over 76 finite official steps, plus a campaign-stratified bootstrap sensitivity. Timeline and transition endpoints use campaign-cluster sensitivity intervals. OpTC has only three top-level campaigns; the report does not treat 76 steps as 76 independent campaigns and does not claim statistical significance when the paired interval crosses zero.

Cost

All five responses were collected for paired counterfactual evaluation. Method costs count only the responses that the frozen policy would query, using recorded tokens and a frozen public-price snapshot. Thus \$22.66 versus \$74.03 is a deployment-policy comparison, separate from the historical expense of collecting the full matrix.

Claim boundary

The supported claim is that CaMVo preserves the complete panel's official-step, timeline, and transition point estimates while reducing replay-policy cost by 69.4%; in auxiliary detection, CaMVo and full voting have the two highest balanced-accuracy point estimates. DeepSeek V4 Pro leads isolated official-step recall by two steps, and this is reported transparently. Results establish performance on the three available OpTC campaigns, not universal campaign-level generalization.

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